# Incident report analysis

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| **Summary** | The organization experienced a **Denial-of-Service (DoS) attack** that disrupted its internal network operations for approximately two hours. During the incident, the network became unresponsive due to a **large volume of incoming ICMP packets** flooding the system. As a result, normal internal traffic was unable to access network resources and services.<br><br>The incident response team acted quickly by **blocking incoming ICMP packets**, temporarily **shutting down non-critical network services**, and prioritizing the restoration of **critical network services** to resume business operations.<br><br>After investigating the event, the cybersecurity team determined that the attack was caused by a **misconfigured firewall**, which allowed an attacker to send excessive ICMP ping requests into the network. This vulnerability enabled the attacker to overwhelm the network and cause the denial-of-service condition.<br><br>To prevent similar incidents in the future, the organization implemented several security improvements. These included creating a **firewall rule to limit the rate of incoming ICMP packets**, enabling **source IP address verification to detect spoofed IP addresses**, deploying **network monitoring software to identify abnormal traffic patterns**, and implementing an **intrusion detection and prevention system (IDS/IPS)** to filter suspicious ICMP traffic. |

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| Identify | The organization's **internal network infrastructure** is a critical asset that supports business operations and employee access to network resources. During the incident, the network became unavailable due to a **Denial-of-Service (DoS) attack** caused by a large volume of incoming ICMP packets flooding the network. This prevented normal internal network traffic from accessing essential network resources.<br><br>The cybersecurity investigation determined that the attack exploited a **misconfigured firewall**, which allowed excessive ICMP traffic to enter the network without restriction. This vulnerability enabled a malicious actor to overwhelm the network with ICMP ping requests, leading to the service disruption. IT administrators and cybersecurity personnel are responsible for managing and securing the network infrastructure, while employees depend on these systems to perform daily business functions. |
| Protect | To prevent similar incidents in the future, the network security team implemented several security controls. A new firewall rule was configured to limit the rate of incoming ICMP packets in order to prevent traffic flooding. The firewall was also updated to perform source IP address verification to detect and block spoofed IP addresses associated with malicious ICMP traffic. Additionally, network monitoring software was deployed to monitor traffic patterns and help identify abnormal network activity. The organization also implemented an intrusion detection and prevention system (IDS/IPS) to analyze incoming traffic and filter suspicious ICMP packets before they reach the internal network. These security controls help strengthen the organization's network defenses and reduce the risk of future denial-of-service attacks. |

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| Detect | To improve threat detection capabilities, the organization implemented network monitoring software to continuously monitor internal network traffic and identify abnormal traffic patterns. This monitoring system helps security personnel detect unusual spikes in network activity, such as excessive ICMP traffic that may indicate a denial-of-service attack. Additionally, the organization deployed an intrusion detection and prevention system (IDS/IPS) to analyze incoming network traffic and identify suspicious characteristics associated with malicious activity. These tools allow the cybersecurity team to quickly detect potential threats and investigate unusual network behavior before it significantly impacts network operations. |
| Respond | During the incident, the organization's incident management team took immediate action to contain and mitigate the attack. The team blocked incoming ICMP packets to stop the flood of malicious traffic entering the network. They also temporarily shut down non-critical network services to reduce network load and prevent further disruption. After containing the attack, the team prioritized restoring critical network services to resume normal business operations and ensure that essential resources were accessible to employees. |
| Recover | After the incident was contained, the incident management team worked to restore normal network operations. Critical network services were restored first to ensure business continuity, followed by the restoration of non-critical network services once the network was stabilized.<br><br>Following the recovery process, the cybersecurity and network security teams implemented several improvements to strengthen the organization's network security. These measures included configuring a new firewall rule to limit the rate of incoming ICMP packets and enabling source IP address verification to |

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| | detect spoofed IP addresses. The organization also deployed network monitoring software to identify abnormal traffic patterns and implemented an intrusion detection and prevention system (IDS/IPS) to analyze and filter suspicious network traffic. These improvements help enhance the organization's ability to recover from similar incidents and reduce the likelihood of future denial-of-service attacks. |

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**Reflections/Notes:**

This incident demonstrates how a simple misconfiguration in a firewall can create a significant security vulnerability that allows attackers to disrupt network operations. The DoS attack highlighted the importance of proper firewall configuration, continuous network monitoring, and layered security controls. Implementing tools such as IDS/IPS and traffic monitoring improves the organization's ability to detect and respond to abnormal network activity. This event also emphasizes the importance of proactive security practices to prevent similar attacks in the future.